

AMINA EMAD

AI & Data Student | Data Analysis & Python | Machine Learning | SQL · Pandas · NumPy
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PROFESSIONAL SUMMARY

Analytical Artificial Intelligence student with hands-on experience in Python, Pandas, NumPy, and Scikit-learn for data analysis and machine learning workflows. Proficient in SQL for structured data querying, exploratory data analysis (EDA), data cleaning, and statistical modeling across real-world datasets. Experienced in designing end-to-end ML pipelines — from raw data preprocessing and feature engineering through model training and performance evaluation. Seeking a Data Analyst Intern or Junior Analyst role to contribute quantitative analysis skills to business intelligence, data-driven decision-making, and actionable insight generation.

TECHNICAL SKILLS

Data Analysis & Python: Python, Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Wrangling, Statistical Analysis, Feature Engineering

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, Predictive Modeling, Model Evaluation, Pattern Recognition, Classification, Regression

Database & Querying: SQL, Structured Data Querying, Data Aggregation, Relational Data Management

Data Processing: OpenCV, Image Data Processing, Feature Extraction, Unstructured-to-Structured Data Transformation, Jupyter Notebook

Tools & Platforms: Python (Automation & Scripting), Git/GitHub, Linux, Docker (Basic), AWS (Cloud Practitioner), Raspberry Pi (IoT Data Collection)

EDUCATION

Bachelor of Artificial Intelligence (In Progress)

Expected: 2027

Faculty of Computers and Artificial Intelligence

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Data Analysis, Embedded Systems, Data Structures & Algorithms

PROJECTS

End-to-End ML Data Pipeline & Classifier | Python · NumPy · Scikit-learn · Deep Learning

- Built a complete machine learning data pipeline processing 70,000+ samples; performed EDA, data preprocessing, normalization, and feature analysis before training a deep neural network to high classification accuracy.
- Evaluated model performance through systematic cross-validation and quantitative metrics analysis, demonstrating structured analytical methodology and iterative model refinement skills.

Multi-Sensor IoT Data Collection & Analysis System | Raspberry Pi · ESP32 · Arduino · Python

- Designed and deployed a real-time multi-variate sensor data pipeline using Raspberry Pi and ESP32; collected, logged, and processed continuous data streams to detect patterns and anomalies in environmental data.
- Implemented threshold-based statistical analysis on live sensor feeds with automated event-driven outputs — demonstrating applied time-series data processing and real-world data collection experience.

Image Data Feature Engineering & Classification Pipeline | Python · OpenCV · Computer Vision

- Developed an end-to-end image data processing pipeline using Python and OpenCV; applied feature extraction and pattern classification to large visual datasets, demonstrating skills in unstructured data transformation.
- Transformed raw image data into structured feature representations for ML classification, directly showcasing data cleaning, feature engineering, and preprocessing workflows applicable to analyst roles.

Smart City Multi-Source Data Analytics Platform | Team Project · Machine Learning · Python

- Collaborated on a data-driven smart city platform integrating heterogeneous IoT sensor datasets with ML models; applied statistical analysis and data fusion techniques to extract actionable insights from real-world data streams.

RESEARCH EXPERIENCE

Co-Author — AI Research Paper | IEEE-Affiliated Conference Submission

- Co-authored and submitted a research paper to an IEEE-affiliated venue; conducted data-driven literature analysis, structured empirical findings, and contributed to technical writing — demonstrating research methodology and quantitative communication skills.

CERTIFICATIONS

Machine Learning Specialization – Coursera (Andrew Ng) | Python for Everybody – University of Michigan (Coursera)

AWS Cloud Practitioner Essentials – Amazon Web Services | Introduction to Cybersecurity – Cisco Networking Academy

LANGUAGES

Arabic (Native) | **English** (Upper Intermediate – Professional Communication)